Cheatsheet AZ-104 — 12 · Virtual networks: VNet, subnets, public IPs, peering, routes and troubleshooting

1. VNet and subnets

Concept	Rules
VNet	Lives in **one region and one subscription**; one or more **address spaces** (RFC 1918 or public; they can be added, expanded or removed **even with peering**, syncing afterwards); limits: 65,536 private IPs, 3,000 subnets, **650 peerings** per VNet (previously 500), 1,000 VNets per region and subscription
Subnet	Cannot overlap or span two VNets; minimum **/29** (8 addresses, **3 usable**): Azure reserves **5** per subnet (network, gateway .1, two DNS .2 and .3, broadcast); a subnet can carry an **NSG**, **route table**, **service endpoints**, **delegation** to a service (App Service, ACI, PostgreSQL...), **NAT Gateway** and the **private subnet** setting (no implicit outbound access)
Special subnets	`GatewaySubnet` (VPN / ExpressRoute, **/27 or larger**), `AzureBastionSubnet` (**/26 or larger**), `AzureFirewallSubnet` (/26), `AzureFirewallManagementSubnet`, `RouteServerSubnet` (/27): exact name and no other resources
VNet DNS	Default: **Azure-provided** (168.63.129.16; resolves VM names in the **same VNet** and public names) · **Custom**: your own servers (DC, BIND); the change requires **restarting or renewing DHCP** on the VMs; can be overridden per NIC. Names across VNets → Private DNS zone (cheatsheet 14)
Inside the VNet	All subnets communicate by default (system route); filtered with NSG; a NIC can only be in a subnet of **its** VNet and **region**

2. IP addresses and NICs

Type	Options	Exam details
Private IP	**Dynamic** (default; kept as long as the NIC exists) or **static** (reserve for DCs, DNS, NVAs)	Assigned from the subnet; configured **on the NIC** (ipconfig), never inside the OS
Standard public IP	**Static** only; **zone-redundant** or zonal; **secure by default** (without an NSG allowing it, nothing gets in); 99.99 % SLA; required with Standard Load Balancer, Bastion, NAT Gateway, Firewall	The **Basic** IP (dynamic or static, no zones) **retired on September 30, 2025**: any new answer is Standard
Public IP prefix	Reserved contiguous block (/28...)	Whitelists at third parties
DNS label	`<label>.<region>.cloudapp.azure.com` on the public IP	"VM reachable by name without Azure DNS"
NIC	Several **ipconfigs** (one primary), several NICs per VM (depending on size; one primary), **IP forwarding** (mandatory on NVAs), **accelerated networking**, own NSG, own DNS	**Deallocating** a VM loses the dynamic public IP (no longer exists with Standard)
Internet egress	**Default outbound access** (implicit Microsoft IP) is going away: new VNets create **private** subnets by default (portal already; API from 2026). **Explicit** egress: **NAT Gateway** (recommended, fixed IP, avoids SNAT exhaustion), public IP on the NIC, **outbound rules** of a public Standard Load Balancer, or NVA / Firewall with UDR	Flexible VMSS and VMs in private subnets **have no egress** without one of these. "VM cannot reach Windows Update / activate" in a private subnet → NAT Gateway
IPv6	Dual stack in VNet, subnet, NIC, Standard LB	Rare questions

3. Peering

Setting	What it does	When
Allow access to remote virtual network	Allows traffic between the two VNets (default yes)	Disable to cut traffic without deleting the peering
Allow forwarded traffic	Accepts traffic that **did not originate** in the remote VNet (e.g. forwarded by an **NVA** in the hub)	Hub-spoke with a firewall / NVA in the hub
Allow gateway transit (on the hub)	The hub offers its **VPN / ExpressRoute gateway** to the spokes	Spokes reach on-prem through the hub gateway
Use remote gateways (on the spoke)	The spoke uses the hub gateway; **cannot have its own gateway** and can only use **one**	Same; also valid with **global** peering

- **Regional** (same region) or **global** (different regions); both over the backbone, no gateway or encryption, latency like inside a VNet; cost per GB. Global: the frontend of a **Basic** ILB cannot be reached and some services do not support it.
- **Not transitive**: A ↔ B and B ↔ C does not give A ↔ C → direct A ↔ C peering, or NVA / Azure Firewall in B with **UDRs** in A and C and *forwarded traffic*. Peering routes show up with next hop **Virtual network peering** in the effective routes.
- Requirements: **non-overlapping** address spaces, create the peering **on both VNets** (the portal does both at once), **Network Contributor** role (or `virtualNetworks/virtualNetworkPeerings/write` + `peer/action`) on both; works across subscriptions and **tenants** (with a guest user or permission on both). States: **Initiated** (the other side is missing) → **Connected**; **Disconnected** if one is deleted.
- Alternative **VNet-to-VNet VPN**: encrypted, transitive via gateways, bandwidth limited by the gateway SKU; only when encryption is required or peering is not possible. For hundreds of spokes → **Azure Virtual Network Manager** (connectivity configurations, up to 1,000 spokes).

4. Routing

Route	Prefix	Next hop	Who creates it
Inside the VNet	Each address space	**Virtual network**	System
Egress	0.0.0.0/0	**Internet** (Azure services do not leave the backbone)	System

Route	Prefix	Next hop	Who creates it
Private RFC 1918 / 100.64	10/8, 172.16/12, 192.168/16, 100.64/10	**None** (dropped) unless they are in the address space	System
Peering	Remote address space	**Virtual network peering**	When the peering is created
On-prem	BGP prefixes or local network gateway prefixes	**Virtual network gateway**	When the gateway is created
Service endpoint	Service IPs	**VirtualNetworkServiceEndpoint** (cannot be overridden)	When the endpoint is enabled on the subnet

- **UDR** (route table associated with one or more subnets; **one** table per subnet; **400** routes per table, 1,000 with Virtual Network Manager): next hops **Virtual appliance** (+ private IP of the NVA or of the ILB in front of the NVAs; the NVA needs **IP forwarding** on its NIC and must live in **another subnet**), **Virtual network gateway** (VPN only, not ExpressRoute), **Virtual network**, **Internet**, **None** (black hole). **Service tags** are also accepted as prefix (e.g. `Storage`).
- **Selection: longest** prefix wins; same prefix: **UDR > BGP > system**. VNet, peering and service endpoint routes take precedence even if there is a more specific BGP route.
- **Propagate gateway routes = No** on the table: the gateway's BGP / static routes **do not** enter that subnet (to force on-prem traffic through an NVA). **Never** disable it on `GatewaySubnet` or associate with it a table that contains 0.0.0.0/0.
- **Overriding 0.0.0.0/0** (forced tunneling) toward an NVA / Firewall / VPN: all traffic leaves that way, including Azure public services without a service endpoint; resources stop being directly reachable from the internet; in private subnets, routes with next hop **Internet** stop working without explicit egress.
- Diagnostics: NIC → **Effective routes** (the VM must be **running**) and Network Watcher → **Next hop**.

5. Connectivity troubleshooting

Tool (Network Watcher)	Answers	Detail
IP flow verify	Does the NSG allow or deny this flow (source, destination, port, protocol, direction)? And **which rule**	First step for "VM1 cannot RDP to VM2"
NSG diagnostics	Evaluation of **all** NSGs in the path for a flow	More complete than IP flow verify
Effective security rules / **Effective routes**	Rules and routes that actually apply to a NIC (subnet + NIC NSG; UDR + system)	On the NIC in the portal, `az network nic list-effective-nsg`, `az network nic show-effective-route-table`
Next hop	Which way traffic to a destination leaves from a NIC	Detects a misplaced UDR or a broken peering
Connection troubleshoot	**One-off** test VM → destination (IP, FQDN, URI, VM): reachable, latency, hops and where it breaks	Needs the Network Watcher extension on the source VM
Connection monitor	Same but **continuous**, with metrics and alerts (cheatsheet 15)	—
Packet capture	pcap from the VM to storage or local, with filters and time limit	Extension required
Topology	Diagram of the VNet and its resources	—
VPN troubleshoot	Diagnostics for VPN gateways and connections	—
VNet flow logs / Traffic analytics	Log of accepted and denied flows (**NSG flow logs** retire in Sept. 2027)	"which traffic is being blocked"

- Checklist when two VMs cannot see each other: **peering** status, **NSG** on NIC and subnet (both must allow), **UDR** (next hop None or NVA down), **OS firewall** (Windows blocks ICMP by default), **DNS** (name vs IP), VM powered off, and nobody blocking **168.63.129.16**.

6. Commands that show up

Task	PowerShell (Az.Network)	Azure CLI
Create VNet and subnet	`$sub = New-AzVirtualNetworkSubnetConfig -Name sub1 -AddressPrefix 10.0.1.0/24; New-AzVirtualNetwork -Name vnet1 -ResourceGroupName RG1 -Location westeurope -AddressPrefix 10.0.0.0/16 -Subnet $sub` · add: `Add-AzVirtualNetworkSubnetConfig ... ; Set-AzVirtualNetwork`	`az network vnet create -g RG1 -n vnet1 --address-prefixes 10.0.0.0/16 --subnet-name sub1 --subnet-prefixes 10.0.1.0/24` · `az network vnet subnet create -g RG1 --vnet-name vnet1 -n sub2 --address-prefixes 10.0.2.0/24`
Subnet: NSG, table, delegation, private	`Set-AzVirtualNetworkSubnetConfig -VirtualNetwork $vnet -Name sub1 -AddressPrefix ... -NetworkSecurityGroup $nsg -RouteTable $rt -Delegation $d; Set-AzVirtualNetwork`	`az network vnet subnet update -g RG1 --vnet-name vnet1 -n sub1 --network-security-group nsg1 --route-table rt1 --delegations Microsoft.Web/serverFarms --default-outbound false`
Public IP and NIC	`New-AzPublicIpAddress -Name pip1 -ResourceGroupName RG1 -Location westeurope -Sku Standard -AllocationMethod Static -Zone 1,2,3` · `New-AzNetworkInterface -Name nic1 ... -SubnetId <id> -PublicIpAddressId <id> -EnableIPForwarding` · static private IP: `$nic.IpConfigurations[0].PrivateIpAllocationMethod = "Static"; Set-AzNetworkInterface`	`az network public-ip create -g RG1 -n pip1 --sku Standard --allocation-method Static --zone 1 2 3 --dns-name app1` · `az network nic create -g RG1 -n nic1 --vnet-name vnet1 --subnet sub1 --public-ip-address pip1 --ip-forwarding true` · `az network nic ip-config update -g RG1 --nic-name nic1 -n ipconfig1 --private-ip-address 10.0.1.10`

Task	PowerShell (Az.Network)	Azure CLI
Peering	`Add-AzVirtualNetworkPeering -Name hub-to-spoke -VirtualNetwork $hub -RemoteVirtualNetworkId $spoke.Id -AllowGatewayTransit` + `Add-AzVirtualNetworkPeering -Name spoke-to-hub -VirtualNetwork $spoke -RemoteVirtualNetworkId $hub.Id -UseRemoteGateways -AllowForwardedTraffic` · `Get-AzVirtualNetworkPeering`	`az network vnet peering create -g RG1 -n hub-to-spoke --vnet-name hub --remote-vnet spoke --allow-vnet-access --allow-gateway-transit` + `az network vnet peering create -g RG1 -n spoke-to-hub --vnet-name spoke --remote-vnet hub --allow-vnet-access --allow-forwarded-traffic --use-remote-gateways` · `az network vnet peering list` · `az network vnet peering sync`
Routes	`New-AzRouteTable -Name rt1 -ResourceGroupName RG1 -Location westeurope -DisableBgpRoutePropagation` · `Add-AzRouteConfig -RouteTable $rt -Name toNVA -AddressPrefix 0.0.0.0/0 -NextHopType VirtualAppliance -NextHopIpAddress 10.0.100.4; Set-AzRouteTable`	`az network route-table create -g RG1 -n rt1 --disable-bgp-route-propagation true` · `az network route-table route create -g RG1 --route-table-name rt1 -n toNVA --address-prefix 0.0.0.0/0 --next-hop-type VirtualAppliance --next-hop-ip-address 10.0.100.4` · associate: `az network vnet subnet update ... --route-table rt1`
NAT Gateway	`New-AzNatGateway -Name nat1 -ResourceGroupName RG1 -Location westeurope -Sku Standard -PublicIpAddress $pip` + subnet `-NatGateway`	`az network nat gateway create -g RG1 -n nat1 --public-ip-addresses pip-nat --idle-timeout 10` · `az network vnet subnet update ... --nat-gateway nat1`
Diagnostics	`Get-AzEffectiveRouteTable -NetworkInterfaceName nic1 -ResourceGroupName RG1` · `Get-AzEffectiveNetworkSecurityGroup` · `Test-AzNetworkWatcherIPFlow -NetworkWatcher $nw -TargetVirtualMachineId <id> -Direction Inbound -Protocol TCP -LocalIPAddress 10.0.1.4 -LocalPort 3389 -RemoteIPAddress 203.0.113.5 -RemotePort 60000` · `Get-AzNetworkWatcherNextHop`	`az network nic show-effective-route-table -g RG1 -n nic1 -o table` · `az network nic list-effective-nsg -g RG1 -n nic1` · `az network watcher test-ip-flow --vm vm1 -g RG1 --direction Inbound --protocol TCP --local 10.0.1.4:3389 --remote 203.0.113.5:60000` · `az network watcher show-next-hop --vm vm1 -g RG1 --source-ip 10.0.1.4 --dest-ip 10.1.0.4`

7. Decision tree

```
What does the wording ask for?
├─ Subnet for N VMs ..... Prefix with N + 5 addresses (minimum /29 = 3 usable)
├─ Connect two VNets, low latency, no encryption ..... Peering (regional or global)
├─ Spokes using the hub VPN / ExpressRoute ..... Gateway transit (hub) + Use remote gateways (spoke)
├─ Spoke A → spoke C through the hub ..... UDR to the hub NVA + Allow forwarded traffic
├─ Encrypted connection between VNets ..... VNet-to-VNet VPN
├─ All outbound traffic through a firewall ..... UDR 0.0.0.0/0 → Virtual appliance (IP forwarding on the NVA)
├─ Keep on-prem routes out of a subnet ..... Route table with "Propagate gateway routes = No"
├─ Drop traffic to a range ..... Route with next hop None
├─ Find out why VM1 cannot reach VM2 ..... IP flow verify → Next hop → Effective routes / rules
├─ Find out which NSG blocks ..... NSG diagnostics / Effective security rules
├─ Fixed outbound IP for all VMs ..... NAT Gateway
├─ VMs with no internet egress in a new VNet ..... Private subnet by default → NAT Gateway / PIP / LB outbound
├─ Public IP that survives zone failures and has an SLA ..... Standard, static, zone-redundant
└─ Two VNets with the same range must talk ..... Impossible with peering: change ranges (or NAT on an NVA)
```

8. Signals → answer

Wording	Answer
"subnet /29, only 3 addresses available"	Correct: Azure reserves 5 per subnet
"address spaces overlap", "peering fails"	Change one VNet's range (it can be resized even with peerings)
"VNet1 peered to VNet2, VNet2 peered to VNet3, VNet1 cannot reach VNet3"	Peering is not transitive: peering 1-3 or NVA + UDR
"spoke VMs must reach on-premises through the hub gateway"	Allow gateway transit + Use remote gateways
"traffic from spoke through the firewall in the hub is dropped"	Allow forwarded traffic is missing on the peering (and the UDR)
"peering status Initiated"	The peering on the other VNet still has to be created
"Basic public IP", "dynamic public IP"	Retired: use Standard static
"VM cannot reach the internet after creating a new VNet"	Private subnet: add a NAT Gateway (or PIP / LB outbound)
"same outbound IP for all VMs", "SNAT exhaustion"	NAT Gateway
"route all traffic through the NVA"	UDR 0.0.0.0/0 → Virtual appliance + IP forwarding
"NVA receives traffic but does not forward it"	Enable IP forwarding on the NIC (and in the OS)
"on-premises routes should not apply to subnet X"	Disable BGP route propagation on its route table
"which rule blocks RDP"	IP flow verify
"traffic takes an unexpected path"	Next hop / Effective routes
"effective routes show nothing"	The VM is deallocated: start it
"VMs cannot resolve each other by name"	Different VNets → Private DNS zone linked to both; custom DNS → the server must forward to 168.63.129.16
"DNS server changed but VMs still use the old one"	Restart the VMs / renew DHCP
"NIC in a different region than the VNet"	Impossible: the NIC must be in the VNet's region

9. Mistakes to avoid

1. Calculating hosts without subtracting the 5 reserved addresses, or using /30 or /31 as a subnet.
2. Believing peering is transitive or that it includes the gateway without enabling transit / remote gateways.
3. Setting Use remote gateways on a spoke that already has its own gateway.
4. Forgetting IP forwarding on the NVA or deploying it in the same subnet it routes for (loop).
5. Associating a table with 0.0.0.0/0 to the GatewaySubnet or disabling its BGP propagation.
6. Assuming a UDR can override a service endpoint route.
7. Designing with Basic IPs or implicit internet egress: they no longer exist for new deployments.
8. Looking for the fault in the OS before checking NSG (NIC and subnet), UDR and peering status.